



ELAM GLOBAL

D.A.G.R. MVP PRD

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SUMMARY AND GOALS

SUMMARY

ELAM GLOBAL D.A.G.R. (Distributed Asset Governance & Retrieval) is a cloud-hosted, agent driven data analytics platform. It provides a team of specialized AI Agents (e.g., “Data Architect,” “Data Engineer,” “BI Developer,” “Data Analyst,” “Data Scientist,” “ML Engineer”) that are embedded within a visual, no-code workflow builder. User construct data pipelines by connecting configurable widgets on a drag-and-drop canvas. Behind the scenes, the agents within these widgets write the necessary code, design data systems, and generate insights based on the user’s configuration, eliminating the need for a traditional data team.

The platform is built on a “Core + Pack” architecture. The **D.A.G.R.** core provides the foundational, mandatory services (ingestion, workflow canvas, governance, security). Industry-specific “Packs” are then layered on top, containing pre-built models, data connectors, and dashboards, for a vertical use-case. The agent features (e.g., code-assist inside widgets) use a model-agnostic adapter with IBM® Watsonx.ai® as the default provider.

The Minimal Viable Product (MVP) will consist of the **D.A.G.R.** core platform and a focused **Risk** analytics pack for calculating portfolio returns and VaR/CvaR. The initial target market is boutique and mid-market investment firms (< USD 150M AUM) in Vancouver, BC, Canada that need automated portfolio analysis but lack the budget for a dedicated data/analytics team.

The primary hosting region will be IBM® Cloud MZR Toronto (ca - tor). For resilience, we will run multiple regional instances within Canada, with IBM® Cloud MZR Montreal (ca - mon) as a warm standby/failover target.

GOALS

The primary goal of the MVP is to deliver a viable, secure, and scalable platform that achieves the following for our initial user base:

User	Capabilities
Analyst	Eliminate 3 – 4 hours of daily manual data preparation, enabling them to ingest portfolio data and generate key risk/return metrics by visually composing analysis pipelines.
Admin	Provide a secure, multi-tenant environment with robust governance (RBAC, audit logs) and system health monitoring, minimizing administrative overhead.
Engineer	Provide the ability to edit and globally save the core logic of analytical widgets, ensuring standardized and powerful tools for all analysts.
Viewer	Access and view pre-built dashboards and download reports for quick view of key insights and analytics.

SUCCESS METRICS

The success of the **D.A.G.R.** MVP will be measure by the following key metrics:

Criteria	Metric(s)
Adoption and Engagement	1. Onboard 5 paying pilot firms by the end of Q4 2025. 2. At least 3 active users per pilot firm. 3. Each pilot firm has at least one production pipeline running on a daily schedule.
Value and ROI	1. 90-day success metric: Achieve $\geq 30\%$ reduction in manual data preparation hours for pilot users. 2. Daily risk reports generated < 15 minutes after market close
Performance and Reliability (SLAs)	1. Availability: $\geq 99.5\%$ monthly uptime for UI and API. 2. Pipeline Latency: End-to-end execution of a typical Portfolio Analytics pipeline (≤ 10GB input) in ≤ 5 minutes. 3. UI Interactivity: P95 page load ≤ 2 seconds.
Financial	1. Gross Margin: $\geq 60\%$ for pilot customers. 2. CAC: $< \text{USD } 5\text{K}$ for initial warm-intro clients.

MILESTONES

SCOPE

D.A.G.R. Core Platform

Function	Criteria
Data Connectors	A suite of connectors that allow users to point to data locations. This includes sources like Google Sheets (via OAuth), Postgres/MySQL databases (via connection string), direct financial data APIs (via user-provided key), and files (CSV/XLSX) located in file storage.
Schema Profiler	Instant data preview, basic stats, and type inference.
Pipeline Builder	Drag-and-drop canvas for visually composing data pipelines by connecting configurable widgets.
AI Agentic Workforce	A team of AI agents (powered by IBM® Watsonx.ai® embedded withing widgets. These agents interpret widget configurations to generate and execute the underlying code (e.g., SQL, Python, etc.) for data transformation, analysis, and visualization.
Execution Engine	DuckDB/Polars-based runner for on-demand and scheduled (cron) pipeline execution.
Governance	Secure, logically isolated workspaces with role-based access control (Viewer, Analyst, Admin, Engineer).
Operations	Ops console with health metrics and failure alerting.
Extensibility	Editable and stored internal widget library.

RISK Pack V1.0

Function	Criteria
Portfolio Returns	Daily time-weighted returns (TWR); MTD/YTD rollups; multi-currency conversion to a portfolio base currency (configurable per-portfolio).
Historical VaR/CVaR	Configurable lookback (default 250d), 95% and 99% confidence levels; CvaR as tail mean; portfolio-level aggregation.
Dashboards	Preconfigured portfolio overview with returns, VaR/CvaR cards, and backtesting exceptions.

Out of Scope*

Function	Criteria
Connectors	Automated, push-based connectors like Bloomberg SFTP and cloud storage event-driven ingestion (e.g., file-drop triggers).
Governance	Column level security, data masking, PII auto-tagging, full data lineage graphs.
Execution Engine	Managed Spark/EMR for very large-scale workloads.
Predictive Models	Predictive analytics, including market drawdown predictors, factor exposure models, and sentiment analysis.
Data Types	Unstructured data (images, text documents, video).
Orchestration	Real-time even stream processing.

REQUIREMENTS

Functional Requirements - General

Function	Criteria
User Personas and JTBD	The platform must serve the primary jobs-to-be-done for the Viewer, Analyst, Admin, and Engineer personas as defined in this document.
Data Ingestion	Users must be able to connect to the specified data sources (CSV, Postgres/MySQL, Google Sheets) with minimal configuration.
Data Profiling	Upon connection, the system must automatically profile the data and present a schema, basic statistics, and data quality metrics.
Pipeline Creation	Analysts must be able to create, save, and modify data pipelines by connecting and configuring widgets on a visual drag-and-drop canvas.
Pipeline Execution	Agent-generated pipelines must be executable on-demand or on a user-defined schedule (cron).
Platform Management	Admins must be able to create workspaces, invite/manage users, and assign roles.
Widget Management	Engineers must be able to edit the base code of widgets and save these changes globally for all users in the organization.

Functional Requirements – **RISK** Pack V1.0

Function	Criteria
Financial Math Conventions	
Holding Period Return (HPR)	Calculated as $\frac{\text{Ending Value} - \text{Beginning Value}}{\text{Beginning Value}}$. For periods longer than a day, this will be calculated by geometrically linking daily returns (i.e., Time-Weighted Return).
Annualized Returns	The geometric average of returns over a period, scaled to a year.
Standard Deviation (Volatility)	The sample standard deviation of daily portfolio returns.

Annualized Volatility	The daily volatility scaled by the square root of the number of trading days in a year (defaulting to 252).
VaR/CvaR	Calculated based on the historical simulation of daily returns. Stored as a positive loss percentage of the portfolio's Net Asset Value (NAV).
Precision	Internal calculations will use float64. Final outputs will be rounded to 4 decimal places for percentages.
VaR/CvaR (Historical Simulation)	
Lookback	250 trading days (configurable from 100-1000).
Levels	95% and 99% confidence.
Construction	Apply historical return shocks per instrument; full re-evaluation at the security level; linear aggregation by current weights.
Outputs	VaR, CvaR (rounded to 2 decimal places), 30d backtest exceptions (Kupiec count), and coverage ratio.
Performance Target	≤ 5 minutes for portfolios up to ~2000 positions with 10 years of history on a bx2-4X16 class worker.
FX and Currency Policy	
Directionality	The rate is defined as `quote_ccy` per 1 `base_ccy` (e.g., `USD/CAD = 1.3500` means 1 USD = 1.35 CAD).
Conversion	Positions and returns are converted to the portfolio's base currency using the T-close FX rate.
Cross Rates	If a direct pair is absent, the system will triangulate via USD if both legs exist; otherwise, it will fail with a `412 - DATA_GAP` error listing the missing legs.
Holidays / Weekends	The FX series is forward-filled to the next business day within the lookback window; price series are not forward-filled.
Governance	Only Admins may change a portfolio's base currency. This change is recorded as immutable metadata for any subsequent runs.

Data Contracts

	Column	Type
Asset Table	date	DATE
	asset_id	STRING
	price	FLOAT
	currency	STRING
	asset_class	STRING
	asset_type	STRING
	irda_classification	STRING
	internal_classification	STRING
	security_type	STRING
	security_name	STRING
	rating	STRING

	Column	Type
Positions Table	date	DATE
	portfolio_id	STRING
	asset_id	STRING
	currency	STRING
	notional_value	FLOAT
	face_value	FLOAT
	purchase_cost	FLOAT
	book_value	FLOAT
	clean_market_value	FLOAT
	security_units	FLOAT

FX Rates Table

	Table	PRIMARY KEY
Primary Keys / Uniqueness	Asset Table	date, asset_id
	Portfolio Table	date, portfolio_id, asset_id
	FX Rates Table	date, base_ccy, quote_ccy

Data Freshness and Corporate Actions

- If latest prices or FX are older than T-2 relative to the run date, the pipeline will fail with a descriptive error.
- The system will require adjusted close prices and warn users if an unadjusted price series is detected.

AI Usage (Agent Features)

Provider	Default to IBM® watsonx.ai® via central adapter
Data Handling	Send only schemas and summaries to agent (no raw data rows); PII redaction enabled. Feature can be disabled per workspace

Non-Functional Requirements

Function	Criteria
Schema Evolution	Backward-compatible changes (e.g., adding a nullable column) are allowed. Breaking changes (e.g., type change, rename, drop) will cause the pipeline to fail with a `422-SCHEMA_INCOMPATIBLE` error, listing the offending columns.
Idempotent Ingest	Uploads will be keyed by `(workspace_id, source_id, file_hash)`. Re-uploading a file with the same hash will be a no-op.
Deduplication	For price and FX data, a unique key per day is enforced. Duplicates are dropped, and a `dq_dedup_events_total` metric is emitted.
Missing Data	If any required price or FX data is missing within a VaR lookback window, the job will fail with a `412-DATA_GAP` error and a report of the missing date ranges.
Bad Prices	If any required `adj_close` is non-positive or NaN within the required window, the job fails with `412-BAD_PRICE` and includes a list of (symbol, date) pairs.
Bad FX Rates	If any required FX `rate` is non-positive/NaN or the pair direction is invalid, the job fails with `412-BAD_FX_RATE`.
Corporate Actions	The system requires adjusted close prices. If an unadjusted series is detected (e.g., split ratio drift $\neq 1$), the job will fail with a `412-UNADJUSTED_SERIES` error.
Time and Calendars	All dates are treated as market days in UTC. For FX data only, holidays and weekends will be forward filled.

Numeric Precision	Returns are stored at 6 decimal places; VaR/CVaR at 2 decimal places. Internal calculations use float64; rounding is applied only on the final write.
Data Residency	All customer data and backups must be stored within Canada. The primary region is IBM Cloud Toronto (ca - tor) with a Canadian failover region IBM Cloud Montreal (ca - mon).
Security	Encryption at rest (AES-256) and in transit (TLS 1.2+). Authentication and authorization handled by IBM Cloud App ID. Immutable audit logs for all sensitive actions.
Scalability	The MVP architecture must support up to 10 tenants, each with up to 10GB of active data, with a design for linear scaling to 5x that capacity.
Availability	Target $\geq 99.5\%$ uptime per region. The failover design supports service continuity across Canadian regions, proven by a quarterly DR test ($RPO \leq 4h$ / $RTO \leq 60m$).
Cost	Infrastructure COGS must be $\leq$ USD 1500 per organization per month at pilot scale.

Acceptance Criteria

Function	Criteria
Performance and Risk Analytics	- Input data contracts (Asset and Portfolio tables) are validated for schema compliance before processing. - Holding Period Return (HPR) for a single day matches the simple return calculation within $1e-8$ tolerance. - HPR for a multi-day period correctly matches the geometric linking of daily returns. - Annualized Return and Annualized Volatility are correctly calculated based on the specified formulas. - Runtime for all performance and risk metrics (excluding VaR/CVaR) for a 2,000-position portfolio over 3 years is ≤ 2 minutes.
VaR/CvaR (Historical Simulation)	- Lookback defaults to 250d; is configurable from 100–1000; invalid values are rejected. - VaR (95,99) equals the empirical quantile of simulated P&L; CVaR equals the mean of losses beyond VaR within $1e-6$ tolerance. 30-day Kupiec exceptions are reported; coverage outside the $[0.5\times, 1.5\times]$ expected band is flagged as a warning. - Runtime: ≤ 5 min for 2,000 positions $\times$ 10y history (bx2-4x16 baseline). - Determinism: identical inputs yield identical VaR/CVaR results.

Core Feature Acceptance

Connectors	• File size limits are enforced (≤ 200 MB via browser). • The profiler completes in ≤ 30 seconds on a 200 MB file and returns types, null percentages, min/max, and sample rows. • Connection credentials are encrypted and stored in IBM Cloud Secrets Manager. • Import jobs are resumable, and partial failures will surface a row-level error sample. • The Bloomberg API connector successfully retrieves daily price data using a user-provided API key.
Scheduler and Execution	• The cron validator rejects invalid schedules. • The number of active runs is limited by the <code>`concurrency_pool`</code> entitlement; excess jobs are queued FIFO. • Each run has an immutable <code>`run_id`</code>, timestamps, status, and artifact pointers. • Failed jobs can be retried manually, which preserves the original inputs for deterministic reproduction.
Governance / RBAC	• Route-level checks enforce permissions for Viewer, Analyst, Engineer, and Admin roles. • Analysts cannot change a portfolio's base currency; this is an Admin-only action. • All sensitive actions (e.g., role change, schedule modification) generate immutable <code>`AuditLog`</code> entries.

USER INTERACTION AND DESIGN

Accessibility and Localization	
Criteria	Requirement
WCAG 2.2 AA	The UI must meet WCAG 2.2 Level AA standards, including full keyboard navigation, visible focus states, color contrast ratios $\geq 4.5:1$, and appropriate labels for all form controls.
Locale	Currency and number formatting will be handled via the workspace locale setting (default `en-US`). All dates will be stored in ISO format (UTC) and displayed with humanized tooltips in the user's local time.
Touch Targets	Minimum interactive target size of 44 x 44 px for all clickable controls.
The user interface will be built in React/TypeScript and will utilize the IBM® Carbon Design System for components, styling, and accessibility. The UI will be organized around nine main views	

LOGIN / SSO GATEWAY

WORKSPACE DASHBAORD

PIPELINE CANVAS

DATA SOURCES

JOB MONITOR

PACK REGISTRY

USER AND RBAC SETTINGS

WORKSPACE SETTINGS

BILLING AND USAGE

ARCHITECTURE AND IMPLEMENTATION

Tech Stack

Function	Criteria
Cloud Provider	IBM® Cloud Toronto (ca-tor); and IBM® Cloud Montreal (ca-mon) warm standby/failover.
Compute	IBM® Cloud Kubernetes Service (IKS) with Cluster Autoscaler.
Backend	Python (FastAPI)
Data Engine	DuckDB + Polars
Frontend	React + Typescript + Vite + TailwindCSS
Metadata/State	IBM® Cloud Databases for PostgreSQL (HA)
File Storage	IBM® Cloud Object Storage
Authentication	IBM® Cloud App ID
CI/CD	Bitbucket Pipelines → IBM® Cloud Container Registry → Argo CD
Observability	Open-Source option: Prometheus + Grafana Enterprise option: IBM® Instana Observability (possible to use with Partner Plus)

Redundancy and Disaster Recovery

Function	Criteria
Compute	Independent IKS clusters per regions, with configurations synced via Argo CD multi-cluster.
Images and IaC	IBM® Cloud Container Registry replication for Docker Images; Terraform state backed up.
Data	IBM® Cloud Object Storage: Cross-Region Replication enabled for all data buckets. Postgres: Nightly logical backups to IBM® Cloud Object Storage (encrypted).
Traffic	IBM® Cloud Internet Services with global load balancing and health checks. RTO target ≤ 60 minutes, RPO ≤ 4 hours, validated by quarterly DR tests.

Agentic AI Adapter

Function	Criteria
Provider	IBM® watsonx.ai® Granite models
Secrets	`WATSONX_API_KEY` and `WATSON_PROJECT_ID` stored in IBM® Cloud Secrets Manager

Service Accounts and API Tokens

Function	Criteria
Service Tokens	The platform will support organized-scoped, non-human API tokens with explicit scopes (e.g., `read:datasets`, `run:pipelines`).
Management	Tokens will be issued via the Admin UI and must be stored securely by the client. A 90-day rotation reminder could be implemented.
Rate Limits	Each token will have its own rate limit (120 req/min), separate from the user-based limits.
Auditing	All API calls made with a service token will be logged in `AuditLog` with `actor_type: service`.
Revocation	Disabling a token is immediate across all services; in-flight requests complete, new requests return 401.

API and Error/UX Contracts

Function	Criteria
Versioning	the REST API will be versioned under `/api/v1`. Breaking changes will necessitate a new version (e.g., `/api/v2`).
Idempotency	all `POST` requests that create resources (e.g., starting a pipeline run) will require an `Idempotency-Key` header.
Pagination	List endpoints will use cursor-based pagination (`?cursor=...&limit=50`).

Exports	All API calls made with a service token will be logged in `AuditLog` with `actor_type: service`.
Standard Errors	The API will use a standard set of error codes, including: `400-VALIDATION`, `401/403-AUTH`, `409-CONFLICT` (e.g., quota hit), `412-DATA_GAP`, `412-BAD_FX_RATE`, `413-PAYLOAD_TOO_LARGE`, `422-SCHEMA_INCOMPATIBLE`, `429-RATE_LIMIT`, `500/503-SERVER_ERROR`.
UI Behaviour	The UI will translate API errors into human-readable messages with clear remediation hints and links to documentation, never showing raw stack traces.

Output Schemas (RISK Pack)		
Function	Criteria	
Model Versioning	`model_version` uses semantic versioning (MAJOR.MINOR.PATCH). Breaking methodology changes (e.g., VaR construction) bump MAJOR ; parameter default changes bump MINOR ; fixes bump PATCH .	
portfolio_performance_summary	Column as_of_date portfolio_id base_ccy holding_period_returns annualized_return_pct annualized_volatility_pct var_95_pct cvar_95_pct model_version ingest_ts	Type DATE STRING STRING FLOAT(4dp) FLOAT(4dp) FLOAT(4dp) FLOAT(4dp) FLOAT(4dp) STRING TIMESTAMP
position_var_contrib	Column as_of_date portfolio_id symbol percent_contrib_var	Type DATE STRING STRING FLOAT

Commercial Model and Entitlements

Deployment Options

Option	What's Included	Customer Responsibilities	Notes for Engineering
Managed Private Cloud	A dedicated, single tenant environment (VPC, IKS, IBM® Cloud Databases for PostgreSQL). All ops, monitoring, backups, and upgrades are handled by ELAM GLOBAL .	None beyond normal tenant admin	Each customer deployment is a separate, isolated stack. Required robust Terraform/IaC for repeatable provisioning.
Customer-Hosted Private Cloud	Helm charts, images, registry access, and a license key. ELAM GLOBAL software support; the customer runs the infrastructure.	Full ownership of the cloud environment, cluster provisioning, uptime/SLOs, backups, and security controls within their own cloud account.	Entitlements are enforced via a license service.

Seats and Roles

Seat Types	Admin, Engineer, Analyst, Viewer
License Rules	- Engineer/Analyst/Admin are paid seats (entitle scheduling and compute) - Viewers are free and do not add scheduling/concurrency entitlements. - Seat entitlements are pooled per org (see below).

Scheduling and Concurrency Entitlements

Item	Entitlement Rule	Rationale / UX
Active Schedules	2 enabled schedules per paid seat, pooled at org level. (e.g., 3 paid seats = 6 enabled crons across the org.)	Scales with team size; simple to meter.
On-Demand Runs	Unlimited within concurrency pool; rate-limited if the pool is saturated.	Let's analysts experiment with burning schedule allotment.
Concurrency Pool	2 concurrent pipeline runs per paid seat, pooled at org level. Minimum floor = 2.	Provides capacity for both scheduled and on-demand runs, reducing contention for time-sensitive jobs.

Soft Burst	24-hour grace above either pool (schedules or concurrency), then older excess schedule auto-disables; admin nudge in-app + email.	Friendly ramp without surprise hard failures.
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Active Schedule: a saved pipeline with `enabled=true` and a cron.

Concurrent Run: a pipeline execution occupying a worker slot from start until terminal state.

Data Volume, Storage, and Retention

Area	Baseline Entitlement	Enforcement / System Behavior
Hot Data (Active Working Set)	Designed and SLO-backed up to 10 GB per workspace	Over hot cap: UI warnings; jobs still run best-effort but SLOs (≤ 5 min) aren't guaranteed. Admin can move data to cold tier.
Cold Data (IBM® Cloud Object Storage Archive)	No hard cap in MVP.	Outputs > 30 days auto-tier to Archive tier (configurable). Retrieval requests show ETA and restore progress.
Result Retention	job logs 90 days; results tables configurable per pipeline (30/90/365 days).	Deletion policy enforced by lifecycle rules; previews hide expired artifacts.
Backups	Managed: nightly snapshots; Customer cloud: customer run.	DR targets: RPO ≤ 4 h / RTO ≤ 60 m (Managed).

API and UI Rate Limits

Scope	Default Limits	Notes
Per-User UI/API	60 req/min sustained, burst 120 for 60s.	Prevents accidental hammering; 429 with retry-after
Per-Org	600 req/min sustained	Aggregated across users and service tokens.

Agent Fair-User (watsonx.ai® default)

Action	Entitlement	Enforcement
“Generate Plan / Explain” (Widget Assistant)	500 invocations/day per org, pooled.	Counter resets daily; after limit, assistant greys out with tooltip.

Token Budget	Soft cap via provider-side policy; warn at 80%, 100%.	No data rows sent (schema + summaries only).	
Environment and Workspaces			
Item	Entitlement	Notes	
Workspaces	Unlimited per org.	Quotas above govern practical usage.	
Environments	Managed: Shared staging + production namespaces per org. Customer: customer defined.	Promotion rules are defined in the CI/CD section below.	
Support Scope and SLAs (Mapped to Deployment)			
Deployment	Incident Response	Uptime Target	Notes
Managed Cloud	P1 ack ≤ 2h biz hours; P2 next business day.	99.5% per region.	Matches observability/alerting section.
Customer Cloud	Software support only; customer owns infra incidents.	N/A (customer SLO)	ELAM GLOBAL provides upgrade notes and runbooks.
License and Telemetry			
License Key Encodes	seat counts, packs enabled, and AI Agent fair-use caps		
Minimal non-PII telemetry events	`schedule_enabled/disabled`, `job_started/finished`, `concurrency_exhausted`, `llm_invocation`. - All events tagged by `org_id` / `workspace_id` and stored in evidence lake per compliance section.		
In-App Usage Visibility			
User page per org	- Seats (used/available), Active Schedules (used/entitled), Concurrency (current/max), Hot Data per workspace, Agent invocations (today/limit). - Threshold alerts at 80% and 100% for each meter. - Enforcement UX: clear error copy, link to docs, and admin CTA to adjust entitlements.		

Engineering To Do

1. Add license/entitlement middleware to API; expose `/usage` endpoints.
2. Implement org-pooled counters for schedules and concurrency, with 24h soft-bursts.
3. Add per-user/org rate limits (429 + Retry-After), and upload size checks.
4. Wire Usage page and toast warnings at 80%/100%
5. Emit telemetry events to the evidence lake.

Further Implementation Details

Observability and Alerting

Function	Criteria
Golden Signals	Each microservice (API, Executor, etc.) must export metrics for latency (P95, P99), traffic (RPS), and error rate (%).
Alerting Rules	P1 (PagerDuty): `5xx Error Rate > 1% for 5m`, `API P99 Latency > 3s for 5m`, `Queue Depth > 100 for 10m`. P2 (Slack): `Disk/Memory Usage > 85%`, `Job Failure Rate > 10% in 1h`. Dashboards: A default Grafana/Instana dashboard will be created for "System Health" (golden signals) and "Pipeline Performance" (job durations, success rates).

Security and Privacy Baseline

Data Classification	All API calls made with a service token will be logged in `AuditLog` with `actor_type: service`.
Dependency Scanning	Disabling a token is immediate across all services; in-flight requests complete, new requests return 401.
Log Retention	Application logs retained for 90 days. Audit logs retained for 365 days.
AI Agent Data Handling	To ensure maximum data privacy and security for the MVP, no customer data (including prompts, responses, or generated code) will be used for model training. All prompts and responses are logged for a 90-day audit period and are masked for sensitive information like emails, names, and API keys.

Developer Experience (DX)	
Local Setup	The project must include a <code>docker-compose.yml</code> file that allows a developer to spin up the entire stack (backend, frontend, database, queue) with a single command.
Seeding	A script must be provided to seed the local database with sample users, workspaces, and pipeline definitions for testing.
Debugging	The runbooks will include clear instructions on how to attach a debugger to running services and how to simulate a failed pipeline run locally.
Pilot Onboarding and Support	
Onboarding Checklist	A standard checklist will be used for each pilot customer, including: • Initial user setup and role assignment • Guidance on providing data that matched the required schemas (Asset Table, Portfolio Table, and FX Rates Table). • Creation of a seed workspace with a pre-built portfolio analysis pipeline.
Support Channel	A dedicated communications channel will be created for each pilot customer for direct communication with the engineering team.
Success Plan	A 30/60/90-day success plan will be co-authored with each pilot, defining their specific goals and metrics.

OPEN QUESTIONS

Targeted Follow-Ups

Function	Question
AI Agent Evolution and Fine Tuning	• How can we implement a feedback loop where the platform learns from an `Engineer's` correction to AI-generated code (e.g., SQL queries)? • What is the architecture, cost, and security model required to support per-organization fine-tuning of the Granite models?
Widget Development Strategy	• What is the process of prioritizing and developing the next set of widgets (e.g., liquidity risk, fundamental ratios, etc.)?
Testing Strategy for Financial Models	• What is the framework for validating the correctness of financial calculations (HPR, VaR/CvaR)? • Will we use a third-party library for validation, or build an internal test suite with known inputs and expected outputs?
Default Workspace Currency	• What should the default base currency be for a new workspace (e.g., USD)?
Data Security	• What are the necessary architecture and infrastructure requirements to ensure SOC 2, GDPR, and ISO 27001 compliance?

REFERENCE MATERIALS

Resource	Link
IBM® Cloud regions	https://cloud.ibm.com/docs/overview?topic=overview-locations
IBM® Design Language	https://www.ibm.com/design/language/
Orange Data Mining	https://orangedatamining.com/
OpenBB	https://github.com/OpenBB-finance
ELAM GLOBAL Investor Presentation	https://drive.google.com/file/d/1y9ZctPInQRfp8glS7U-Oq_NKoZe8AAQ7/view?usp=drive_link
WCAG 2.2 Standard	https://www.w3.org/TR/WCAG22/